Flow, Autonomy, and Organizational Heat: A Physical Systems Analogy for Psychology, Decision Architecture, and Organization Design

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Abstract

Organizational systems frequently exhibit patterns of friction, conflict, and systemic overload that resist explanation through conventional accounts of motivation, leadership, or communication. This paper develops a conceptual framework, grounded in hydraulic and electrical circuit analogies, to model how autonomy, legitimacy, capability, risk, and structural complexity interact to produce either smooth coordinated action or what we term *organizational heat*—the dissipative byproduct of action flow encountering structural resistance. The framework proposes five core formal relations as modeling hypotheses: autonomy as the product of legitimacy and capability; legitimacy and capability as bounded quantities subject to reduction by insignificance and risk, respectively; action flow as a ratio of autonomy to complexity resistance; and organizational heat as proportional to the square of flow times resistance. It further distinguishes between legitimacy as durable standing within a decision domain and admissibility as the execution-time validity of a specific act under current conditions. These algebraic forms are the authors' own formalizations by analogy to physical circuit theory and are not directly derived from the cited literatures, which are invoked to motivate the conceptual components rather than to validate the specific functional forms. The model is extended through Kirchhoff-like network constraints applied to decision architectures, through a distinction between coordination flow and irreversible commitment, and through analysis of how AI-augmented organizations alter the flow–resistance–heat balance while increasing the importance of commit-time admissibility checks. We conclude with testable predictions that could validate or refute the framework's structural assumptions, and with an explicit assessment of model limitations.

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1 Introduction

1.1 The Problem of Organizational Instability

Organizations routinely experience coordination failures, decision overload, rework, and escalating interpersonal conflict even when their members are individually competent and their objectives are clearly articulated. Standard accounts of such failures frequently invoke motivational deficits (Deci & Ryan, 1985), leadership dysfunction (Yukl, 2012), or communication breakdowns (Weick, 1995). While each of these accounts captures important phenomena, they do not adequately explain a class of instabilities that arise structurally—that is, as a consequence of how decision authority, information load, workflow routing, and interdependency are arranged, rather than as a consequence of individual deficiency or interpersonal failure.

Physical systems provide a potentially productive set of analogies for this structural class of problem. In fluid and electrical networks, flow is governed by driving potential and opposed by resistance; when flow encounters resistance, energy is dissipated as heat (Feynman et al., 1963). This paper proposes that analogous structural constraints may apply to the flow of decisions and actions within organizations. The specific algebraic forms through which we formalize this analogy—presented fully in Sections 2–6—are modeling choices rather than empirical findings; they are introduced because they generate tractable, falsifiable predictions, not because they are established by the literatures cited for motivation.

A further refinement is necessary. In complex organizations, the first visible symptoms of failure often appear in execution—for example, on the shop floor, in service delivery, in customer outcomes, or in compliance breaches. Yet execution commonly reveals the symptom rather than the point of origin. Likewise, data inconsistencies, reporting anomalies, or KPI conflicts may appear to be the first breakdown, while in fact they often reflect a prior distortion in how the underlying work was decomposed, routed, sequenced, escalated, or reconciled. The present framework therefore gives analytic priority to the *workflow or orchestration layer*: the intermediate structure through which strategic and managerial intent is translated into tasks, handoffs, system actions, escalation paths, and coordinated execution.

This distinction matters because organizations do not act directly on strategy. They act through orchestrated workflows. If the workflow or orchestration layer encodes a slightly altered version of the original decision, then downstream actors, systems, and data structures may all behave in ways that are locally consistent while globally misaligned with the organization's actual intent. The present paper therefore argues that many coordination failures should be understood not simply as failures of communication or authority, but as failures of *intent preservation through orchestration*.

A second complication arises in distributed and partially automated systems: even when decisions are initially well-founded and even when orchestration remains coherent, the conditions that made a specific action valid may change before the action is actually committed. This implies that organizations face not one but two distinct control problems. First, they must preserve coherent intent as decisions are translated into workflows across humans, teams, software systems, and AI agents. Second, they must determine whether a specific act remains admissible at the moment it becomes irreversible. The present paper argues that both problems must be modeled explicitly and that they occur at different structural layers.

1.2 Literature Positioning

The framework draws on five bodies of established theory to motivate its conceptual components. We note explicitly where the framework is aligned with existing evidence and where it extends beyond what those sources directly support.

Across these literatures, authority, control, and validity are often discussed without sharply distinguishing durable standing from execution-time authorization, and workflow structure is often treated as an implementation detail rather than as a primary site of strategic distortion. The present framework departs from this tendency in two ways. First, it distinguishes durable standing legitimacy from execution-time admissibility. Second, it treats workflow or orchestration architecture as the principal intermediate layer at which intent is either preserved or lost before downstream data and execution outcomes make the loss visible.

Self-Determination Theory (SDT). Deci and Ryan's self-determination theory (Deci & Ryan, 2000; Ryan & Deci, 2000) identifies autonomy, competence, and relatedness as fundamental psychological needs whose satisfaction reliably predicts intrinsic motivation and well-being across a wide range of empirical studies. This evidence base motivates our choice of autonomy as the primary driving variable in the model. The present framework departs from SDT in treating autonomy as a structural quantity with a specific algebraic form (the product of legitimacy and capability), rather than as a psychological need whose satisfaction falls on a continuous scale. That algebraic form is a modeling assumption, not something SDT or its associated empirical literature directly implies.

Cognitive Load Theory (CLT). Sweller's cognitive load theory (Sweller, 1988, 1994) establishes that working memory is a finite resource and that performance degrades when total processing demand—across its intrinsic, extraneous, and germane components—exceeds available capacity. This finding supports the general principle that complexity imposes a cost on effective action, which motivates the resistance construct in the present

model. The specific claim that workflow orchestration and organizational complexity function jointly as resistance in a ratio-form flow equation is our own formalization, not a conclusion of CLT.

Systems Theory and Cybernetics. General systems theory (Bertalanffy, 1968) and cybernetics (Ashby, 1956; Wiener, 1948) model organizations as open systems regulated by feedback, characterized by flows of energy, information, and material. This tradition motivates the use of flow and network concepts in describing organizations. The application of Kirchhoff-like conservation constraints to decision networks—developed in Section 7—is our extension of this tradition to the domain of authority and coordination flows; it is a conceptual analogy, not an established result in organizational systems theory.

Control Theory and Normal Accidents. Perrow's normal accident theory (Perrow, 1984) and Beer's viable systems model (Beer, 1972) both establish that organizational failure is systematically produced by structural mismatches between complexity and regulatory capacity. This provides empirical and theoretical grounding for the claim that structural properties, rather than individual behavior alone, determine system stability. The present framework extends this claim by proposing that one important locus of early instability is the workflow or orchestration layer, where regulatory intent is first transformed into operational sequence, routing, and handoff logic. The further claim that such instability manifests as a quadratic heat function remains a model prediction that requires empirical testing against the failure-mode patterns that Perrow and Beer document.

Coordination Costs and Organization Design. Transaction cost economics (Williamson, 1981) and coordination theory (Malone & Crowston, 1994) establish that organizational design choices have systematic effects on the costs of integrating distributed action. Puranam et al. (2012) demonstrate that coordination failure is predictably linked to specific structural properties of task interdependence. The present model builds on this tradition by treating coordination cost not only as a static burden but as a dynamic resistance parameter that interacts with autonomy to produce either stable throughput or dissipative heat. It further argues that in high-complexity systems this burden is mediated centrally by orchestration architecture: the structural layer where decisions are decomposed into executable paths across roles and systems.

1.3 Overview of the Framework

The model proposes the following formal relations and architectural distinctions as a coherent set of modeling hypotheses:

1. Autonomy as the product of legitimacy and capability

2. Legitimacy as a bounded quantity reduced by insignificance
3. Capability as a bounded quantity reduced by risk
4. Action flow as autonomy divided by complexity resistance
5. Organizational heat as the square of flow multiplied by resistance
6. Kirchhoff-like conservation constraints on decision networks
7. Decision architecture as the structural allocation of standing authority
8. Orchestration architecture as the structural layer through which intent is decomposed, routed, sequenced, escalated, and either preserved or distorted
9. Commit architecture as the determinant of whether action remains admissible at the moment of irreversible execution

These relations do not all operate at the same layer. The first six concern the generation, routing, and dissipation of coordinated action through an organizational network. The seventh concerns who possesses recognized standing to decide within a domain. The eighth concerns how those decisions are translated into coordinated workflows across human and technical components. The ninth concerns a distinct but related problem: whether a specific action remains admissible at the point of commitment. Their value as a theoretical contribution depends not on the precision of the analogy to physics but on whether they generate predictions that can be tested against organizational data. Section 12 develops these predictions explicitly.

Figure 1 presents the integrated model.

2 Decision, Orchestration, and Commit Architecture

Decision architecture is defined in this framework as the structural configuration that allocates standing authority across an organizational system: who may decide, within which domain, under which recognized mandate, and with what escalation relationships. In the present model, decision architecture primarily shapes legitimacy and the distribution of recognized decision rights. It answers the question of *who is structurally entitled to decide in principle*.

Decision architecture alone, however, does not determine whether intent survives translation into action. Organizations rarely act directly on decision rights. They act through workflows. For this reason, the present framework distinguishes decision architecture from *orchestration architecture*: the structural configuration through which decisions are decomposed, routed, sequenced, escalated, reconciled, and translated into coordinated

action across human actors, software systems, and AI agents. Orchestration architecture is the primary design lever for $\text{Resistance}_{\text{Complexity}}$ and the principal intermediate layer at which coherence is either preserved or lost.

A further distinction is required. Even when decision rights are clear and workflow orchestration remains coherent, a specific act may still no longer be valid when execution occurs. For that reason, decision architecture and orchestration architecture must both be distinguished from *commit architecture*: the structural mechanism through which admissibility is resolved at the point where action becomes materially, financially, technically, or institutionally irreversible.

We denote execution-time admissibility for a specific act a at time t as

$$\mathcal{A}(a, t) \in \{0, 1\},$$

where $\mathcal{A}(a, t) = 1$ indicates that the act remains valid under current conditions at the commit point, and $\mathcal{A}(a, t) = 0$ indicates that it does not. This notation is intentionally minimal. The present paper does not yet specify the full functional form of admissibility, but introduces it to mark the conceptual distinction between durable standing legitimacy and execution-time validity.

2.1 Relationship to Governance, Management, and Control Systems

The three architectural layers introduced here operate at a different level of abstraction from governance, management, and control systems. Governance specifies who has ultimate authority. Management specifies how that authority is exercised in planning, coordination, and supervision. Control systems specify how deviations from desired states are detected and corrected. The present framework introduces a more structural decomposition.

Decision architecture specifies how standing authority is distributed across roles and domains. Orchestration architecture specifies how those decisions are transformed into routable, sequenced, and enforceable action paths. Commit architecture specifies how the system determines whether a specific act may still proceed when execution becomes irreversible. Together these layers define not merely who governs and who acts, but how intent travels from recognized authority into operational reality and whether that action remains valid when it commits.

This construct is conceptually related to Beer's viable systems model (Beer, 1972), in which system-level cohesion depends on the regulatory structure of the metasystem. It is also related to Williamson's treatment of organizational form as a determinant of transaction costs (Williamson, 1981). The present model extends these frameworks by separating the distribution of authority from the preservation of intent through orchestration and from the execution-time resolution of admissibility. Whether this layered decomposition adds

predictive value over more traditional governance and coordination-cost accounts is an open empirical question.

2.2 Workflow and Orchestration as the First Instability Layer

The framework proposes that in many complex organizations the first operational instability appears in the workflow or orchestration layer rather than in execution itself. The shop floor, customer interface, service outcome, or compliance event typically reveals the symptom, but not the origin. Similarly, data discrepancies, KPI conflicts, or reporting anomalies often reflect already-distorted workflow logic rather than causing it. The orchestration layer is therefore analytically privileged: it is where strategic and managerial intent is first converted into routable, sequenced, and enforceable action logic, and where small reinterpretations can become systemically durable.

This claim has a specific implication. A workflow may remain locally coherent at each step while globally encoding a distorted version of the original decision. Handoffs may be clear, approvals correctly assigned, and local metrics satisfied, yet the overall sequence may pursue a slightly altered objective because the decomposition, routing, sequencing, escalation logic, or reconciliation structure has already shifted the effective meaning of the decision. In such cases, downstream execution is not malfunctioning in the ordinary sense; it is functioning faithfully relative to a workflow that has already drifted.

The framework therefore treats orchestration architecture not as a minor implementation detail but as the principal structural layer through which intent survives or degrades. This is also the layer where human and machine action most directly interact: workflow engines, task routing systems, approval paths, user interfaces, exception handling, vendor handoffs, and agentic AI systems all operate here. As a result, this layer is expected to become increasingly decisive in AI-augmented organizations.

2.3 Commit Architecture and the Point of Irreversible Action

Commit architecture is defined here as the structural arrangement by which the system determines whether a specific act remains admissible at the point where execution becomes materially, financially, technically, or institutionally irreversible. Typical commit points include code deployment, release of funds, grant of access, publication of instructions, execution of a trade, issuance of an approval that triggers downstream automation, or activation of an AI-originated action in a live environment.

This distinction matters because upstream governance artifacts—policies, role assignments, workflow approvals, and documentation—shape and constrain action but do not by themselves constitute control in the strong sense. If the validity of the act is not resolved where the system commits, those upstream elements function as preparation rather

than decisive control. The model therefore treats commit architecture as a separate and necessary complement to both decision architecture and orchestration architecture.

A robust commit architecture must do at least three things. First, it must identify the commit point clearly enough that irreversible action is not occurring invisibly elsewhere in the stack. Second, it must make the admissibility conditions for that class of action explicit enough to be evaluated under current conditions. Third, it must bind the result of that evaluation directly to execution, so that inadmissible actions are blocked rather than merely recorded, questioned, or retrospectively audited.

This concept is especially important in distributed and AI-augmented environments, where the interval between preparation and execution may be short, the number of candidate actions large, and the contextual conditions governing admissibility subject to rapid change.

2.4 Structural Sources of Resistance

The following structural properties are proposed to increase Complexity Resistance, with special emphasis on the orchestration layer as the place where these burdens are most often operationalized:

- **Authority ambiguity:** role conflict and boundary ambiguity are empirically associated with performance decrements and stress (Rizzo et al., 1970); the model interprets this as a resistance increase.
- **Responsibility overlap:** diffusion of accountability is associated with reduced individual effort (Latané et al., 1979); the model treats this as a coordination friction that frequently becomes operationally embedded in workflow handoffs and exception paths.
- **Information–authority separation:** when decision rights are held by agents who lack the specific knowledge required to exercise them, information must be transferred before action can proceed (Jensen & Meckling, 1992); this transfer cost enters as resistance and often appears concretely as orchestration burden.
- **Incentive misalignment:** divergent incentive structures among interacting agents generate negotiation costs at decision nodes (Holmström & Milgrom, 1991), often forcing workflow branches, informal workarounds, or repeated reconciliation.
- **Feedback latency:** delayed feedback is a documented source of oscillation and instability in decision systems (Sterman, 1989); the model treats it as a resistance component that increases the probability of constraint violations and stale orchestration logic.

- **Orchestration distortion:** decomposition logic, routing rules, sequencing assumptions, escalation pathways, and handoff structures may encode a slightly altered version of the original decision. The present framework treats this as a primary source of coherence loss rather than as a downstream symptom.

2.5 Principles of Low-Resistance and High-Integrity Architecture

From the model's assumptions, the following design principles reduce Complexity Resistance and are therefore predicted to improve flow stability, reduce heat, and preserve intent more reliably:

1. Assigning clear, non-overlapping decision authority for each recognized decision class.
2. Distinguishing explicitly between decision rights and the workflow structures through which those rights are operationalized.
3. Collocating authority with the specific knowledge required to exercise it (Jensen & Meckling, 1992).
4. Designing orchestration logic—decomposition, routing, sequencing, escalation, and reconciliation—so that the original decision can be translated into action without silent reinterpretation.
5. Designing feedback mechanisms with sufficiently low latency to support the required throughput rate and to detect workflow drift before it becomes embedded in downstream data and execution.
6. Resolving constraint conflicts at the architectural design level, so that individual decision nodes and workflows receive consistent, satisfiable constraint sets.
7. Limiting the number of simultaneous incoming constraints at any node to within its estimated processing capacity.
8. Making the point of irreversible commitment explicit and binding admissibility checks directly to that point, so that validity is resolved where execution occurs rather than assumed from earlier preparation.

These are predictions of the model. The general principles are consistent with established design guidance (Mintzberg, 1979; Williamson, 1981), but the specific prediction that violating them generates superlinear heat and early orchestration instability—rather than merely linear coordination cost—is a novel claim that requires empirical support.

3 Capability and Risk

$$\text{Capability} = \text{Capability}_{\text{max}} - \text{Risk}$$

Capability denotes the effective ability of an agent to execute actions reliably under prevailing conditions. The framing as a situationally bounded quantity reflects the well-established finding that performance depends not only on skill but on the demands imposed by the environment: the same agent may be highly capable under routine conditions and substantially impaired under conditions of high uncertainty, cognitive load, or consequence severity.

$\text{Capability}_{\text{max}}$ represents the agent's full repertoire of skills, knowledge, and resources under low-demand conditions. Risk in this model is not a property of outcomes per se but a demand on cognitive and attentional resources. This framing draws on Kahneman's model of attentional capacity (Kahneman, 1973), which establishes that higher-demand tasks reduce the resources available for concurrent activities, but extends it to the organizational level in a way that source does not directly imply.

The following factors are proposed as determinants of the Risk demand variable:

- **Environmental uncertainty:** unpredictability increases monitoring demands and error-detection requirements (Perrow, 1984).
- **Task complexity:** higher intrinsic task complexity consumes working memory capacity (Sweller, 1994), leaving less available for productive execution.
- **Consequence severity:** high-stakes decisions direct attention toward loss-avoidance (Kahneman & Tversky, 1979), reducing resources available for efficient action generation.
- **Knowledge gaps:** operating under conditions of bounded rationality (Simon, 1972) increases the cognitive cost of each decision, which the model treats as a reduction in net Capability.

As with the Legitimacy equation, the linear subtraction form is a modeling simplification. The functional relationship between risk factors and effective capability is almost certainly nonlinear in practice, and the specific choice of linear subtraction is made for analytical tractability. The parallel algebraic structure of the Legitimacy and Capability equations is also a deliberate formal symmetry: both components of Autonomy are modeled as bounded resources subject to environmental depletion. Whether this symmetry reflects a deeper empirical regularity or is merely a convenient formal property of the model is an open question.

4 Action Flow

$$F = \frac{\text{Autonomy}}{\text{Resistance}_{\text{Complexity}}}$$

Action flow F represents the rate at which decisions and coordinated actions are generated and completed per unit time within the organizational system. The ratio form is directly inspired by Ohm's law in circuit theory (Feynman et al., 1963), where current equals voltage divided by resistance. This is an analogy, not a derivation: there is no prior theoretical or empirical result that directly implies this functional form for organizational systems. The ratio is proposed because it generates clear, testable predictions about the effects of autonomy expansion and resistance reduction as independent and interacting interventions.

Action flow should not be equated with valid commitment. A decision, instruction, or prepared action may move efficiently through an organizational structure without yet being executable under current conditions. For this reason, the present framework treats flow as the movement of coordinated action potential through the system, while reserving the question of whether a specific act may actually proceed for later treatment under the concept of admissibility at commitment.

$\text{Resistance}_{\text{Complexity}}$ subsumes all structural and informational factors that impede the throughput of decisions and actions. The following categories of resistance are proposed, each supported by evidence that they increase coordination cost or reduce decision throughput:

- **Coordination requirements:** the need to achieve alignment among multiple actors before action can proceed has been established as a systematic source of organizational cost (Malone & Crowston, 1994).
- **Information overload:** processing demand in excess of working memory capacity degrades decision quality and speed (Sweller, 1988).
- **Constraint conflict:** simultaneous obligations that cannot all be satisfied by a single decision force negotiation or escalation, both of which consume resources that would otherwise support throughput.
- **Structural opacity:** ambiguous authority maps and missing escalation paths are documented sources of coordination failure (Mintzberg, 1979).

The ratio form implies that flow can be increased either by increasing Autonomy or by decreasing Resistance. These are architecturally distinct interventions with different cost profiles. However, the critical implication of the ratio form—in combination with the heat equation—is that high Autonomy under high Resistance does not produce high stable flow; it produces high organizational heat (Section 6). Whether this prediction holds

quantitatively, as opposed to qualitatively, is an empirical question the model cannot answer from its assumptions alone.

5 Organizational Heat

$$H = F^2 \times \text{Resistance}$$

Organizational heat H is the central novel construct of this framework. It is defined as the rate of unproductive dissipation within the organizational system—manifesting as stress, interpersonal conflict, rework, escalation, and decision fatigue. The term and its formal expression are direct analogies to Joule heating in electrical systems (Feynman et al., 1963), where power dissipation equals the square of current times resistance. This is a conceptual innovation of the present model and is not a standard construct in any of the literatures cited for motivation.

The quadratic dependence on flow is the most consequential structural feature of the model. Under this formulation, doubling flow while holding resistance constant quadruples heat. This non-linearity generates a specific prediction: systems that operate stably under moderate action rates may exhibit abrupt, disproportionate degradation under higher loads. This qualitative pattern is documented across several literatures. The Yerkes–Dodson relationship between arousal and performance (Yerkes & Dodson, 1908) establishes an inverted-U pattern in which high arousal impairs performance—consistent with the prediction that high flow generates costs—but does not imply a quadratic relationship specifically. Perrow's analysis of tightly coupled systems (Perrow, 1984) documents how small perturbations can cascade to catastrophic failure, which is consistent with superlinear heat accumulation but is not formalized in those terms. The specific quadratic form of the heat equation is a modeling assumption that warrants direct empirical investigation.

This formulation captures dissipation generated by friction in the movement and reconciliation of coordinated action. It does not by itself capture a distinct class of failure in which an action reaches the point of execution efficiently but should not occur because its authorizing conditions are no longer satisfied. Such failures are not best understood as high heat alone, but as failures of admissibility resolution at commitment. The two may interact: high heat increases the likelihood of stale, bypassed, or weakly validated commitments, but the concepts should remain analytically distinct.

Substituting the flow equation yields the alternative expression:

$$H = \frac{\text{Autonomy}^2}{\text{Resistance}_{\text{Complexity}}}$$

This form reveals an important design asymmetry: under the model's assumptions, increasing Autonomy without reducing Resistance raises heat superlinearly, while reducing

Resistance (holding Autonomy constant) reduces heat. The practical implication is that resistance reduction is a more leverage-efficient intervention than autonomy expansion under high-complexity conditions. This is a prediction of the model, not an established finding.

When heat is not dissipated through structural adjustment, the model predicts accumulation as latent instability. The psychological correlates proposed—burnout (Maslach & Jackson, 1981), trust erosion (Mayer et al., 1995), and risk aversion—are all empirically documented outcomes of sustained organizational overload. The model proposes that they share a common structural cause expressible through the heat equation; whether that causal unification is valid is an empirical question.

6 Kirchhoff Constraints in Decision Networks

The scalar relations of Sections 5–6 describe individual agents or units in isolation. Real organizations are networked: decisions depend on prior decisions, authorities overlap, and information flows along paths. To model these network effects, we introduce Kirchhoff-like conservation constraints by direct analogy to Kirchhoff's current and voltage laws in electrical network theory (Feynman et al., 1963).

These constraints are the authors' extension of the flow analogy to networked systems. They are not derived from organizational theory but are proposed as useful formal tools for identifying structural sources of coordination failure. Their value depends on whether the predictions they generate— developed in Section 12—are empirically supported.

6.1 Node Conservation: Flow Balance

We propose that at any decision node n in the organizational network, a conservation-like constraint should hold between incoming and outgoing decision flow:

$$\sum_i F_{\text{in},i} = \sum_j F_{\text{out},j}$$

A decision node represents any locus where multiple information streams, authorities, or constraints must be reconciled before an output decision can be produced. Common examples include a project manager integrating inputs from engineering, finance, and compliance before committing to a delivery plan, or a cross-functional committee reconciling competing resource claims.

The proposed conservation principle states that if incoming flows consistently exceed the node's processing capacity, the excess accumulates as a backlog of unresolved decisions. This backlog is itself a source of increased Resistance for subsequent flow, creating a positive feedback: backlog increases resistance, which reduces effective throughput,

which may increase heat as upstream agents attempt to force decisions through the congested node. This feedback structure is qualitatively consistent with documented patterns of decision escalation and overload in management research (Sterman, 1989), but the formal correspondence has not been empirically established.

Decision nodes must reconcile all incoming constraints simultaneously, which may include strategic goals, resource limits, risk and compliance policies, stakeholder demands, and temporal dependencies. When the incoming constraint set is overdetermined—when no single decision can simultaneously satisfy all constraints—the node cannot produce a valid outgoing flow. The model characterizes this as a node-level conservation violation. In practice, such violations manifest as decision deadlock or escalation; the model predicts that nodes with chronically overdetermined constraint sets will be persistent heat sources. This prediction is testable and is stated formally in Section 12.

6.2 Loop Constraints: Potential Balance

For closed loops within the decision network, we propose the following constraint by analogy to Kirchhoff's voltage law:

$$\sum_{\text{loop}} \text{Autonomy}_k = \sum_{\text{loop}} F_k \times \text{Resistance}_k$$

This constraint states that the total driving potential invested in a decision cycle must be fully accounted for by productive work done against resistance within that cycle. When the balance is violated—when more potential is invested than the cycle can convert to output—the residual accumulates as heat or forces revision of the assumptions underlying the cycle.

Organizational loops subject to potential imbalance include:

- **Strategy–execution–evaluation–strategy cycles:** where strategic ambition is not matched by the capability and resistance profile of the execution layer, producing persistent performance gaps that the feedback mechanism cannot correct (Sterman, 1989).
- **Policy–implementation–reporting–policy cycles:** where compliance reporting is structurally decoupled from implementation realities, producing cycles that systematically miscalibrate the policy layer.
- **Objective–action–feedback–objective cycles:** where feedback latency or distortion prevents accurate potential signals from reaching the goal-setting process.

6.3 Coordination Failures as Conservation Violations

The framework proposes that many coordination failures can be formally characterized as violations of node or loop conservation. A unit that generates more outgoing decisions than its incoming standing authority and constraints can support is operating with a legitimacy or coherence deficit. A unit that absorbs incoming decision flow without producing proportionate outputs is generating a resistance backlog. Both conditions degrade systemic flow efficiency and increase total organizational heat. However, satisfaction of these conservation-like constraints is not sufficient for full control, because a network may remain internally balanced while still committing an inadmissible act if validity is not resolved at the point where execution becomes irreversible.

This characterization has diagnostic value: it suggests that persistent coordination failures can be localized to specific nodes or loops where conservation is violated, and that structural interventions targeting those points should be more effective than diffuse cultural or motivational interventions. Whether this diagnostic strategy outperforms alternatives in practice is an empirical question that the present framework cannot resolve on theoretical grounds alone.

7 AI-Native Organizations

7.1 How AI Augments Potential Flow

AI-augmented organizations—systems in which artificial intelligence performs a substantial fraction of information processing, decision-support, or execution functions—alter the model's parameters in ways that are qualitatively distinctive. There is empirical evidence that AI tools extend the range of tasks that can be executed reliably (Agrawal et al., 2018) and reduce the time required per decision, which in the model's terms increases Autonomy at the agent or unit level by reducing Risk demand on human Capability. Automation similarly reduces variance in routine execution, decreasing the uncertainty component of Risk.

Under the model's flow equation $F = \text{Autonomy}/\text{Resistance}$, these effects produce a substantial increase in potential action flow. Whether that potential is realized as productive output or dissipated as heat depends on whether Resistance is adjusted proportionately. The model's prediction—and this is explicitly a model-derived prediction rather than an observed regularity—is that when AI increases Autonomy without a proportionate reduction in Resistance, the heat equation $H = F^2 \times \text{Resistance}$ predicts superlinear heat accumulation.

7.2 Why Heat May Increase Faster Than Flow

The model predicts that AI-augmented systems face an asymmetric risk: Autonomy-increasing mechanisms and Resistance-reducing mechanisms operate independently and are not automatically co-produced. Several properties of AI systems are proposed to generate resistance of their own:

- **Objective misalignment:** AI systems optimizing proxy metrics can diverge from organizational intent, creating constraint conflicts at integration nodes (Russell, 2019). The model treats each such conflict as a Kirchhoff node violation.
- **Metric proliferation:** multiple AI subsystems generating competing performance signals increase constraint overdetermination at human oversight nodes.
- **Opacity, legitimacy loss, and weak admissibility resolution:** non-interpretable AI decisions may reduce the social legitimacy of AI-generated standing within the human organizational system, while also making it more difficult to determine whether a specific AI-originated act remains admissible under current conditions. The model formalizes the first as an increase in Insignificance for AI-origin flows and treats the second as a commit-architecture problem.
- **Pace asymmetry:** AI subsystems may generate decisions at rates that exceed the processing capacity of human governance nodes, producing backlogs that the model identifies as conservation violations.
- **Cascading dependencies:** tightly coupled AI pipelines increase the topological complexity of the decision network, multiplying the number of Kirchhoff constraints that must be simultaneously satisfied.

These effects are proposed as mechanisms by which AI integration could generate heat faster than flow. Whether they do so in practice, and whether the magnitude follows the quadratic prediction, are empirical questions that organizational research on AI-augmented work could in principle address.

7.3 Architectural Requirements for AI-Augmented Stability

The model generates the following specific structural predictions for AI-augmented organizations:

- Each AI agent requires a clearly specified authority domain, understood as the class of actions or recommendations for which it may generate prepared outputs under normal conditions.

- Irreversible AI-originated actions must be subject to execution-time admissibility checks bound directly to the commit point, rather than inferred solely from prior approvals, static rules, or durable role assignments.
- The objective functions of interacting AI agents must be mutually consistent to avoid loop-level potential imbalance and locally valid but globally misaligned action.
- Human oversight nodes must have sufficient processing capacity to handle the flow generated by AI subsystems, or AI decision rates must be deliberately throttled to match human governance bandwidth.
- Feedback latency must be low enough to detect objective drift, contextual change, and constraint conflict before they accumulate as unresolved backlogs or invalid commitments.

These are architectural prescriptions derived from the model's assumptions. They are consistent with the governance recommendations of Russell (2019) and with emerging frameworks for AI safety and controllability, but their specific derivation from the heat equation is novel to this framework.

In AI-augmented settings, the distinction between standing legitimacy and admissibility becomes operationally decisive. A model may be correctly assigned to a decision domain, yet any particular act it advances toward execution may still be invalid when current constraints, permissions, exposures, or environmental conditions are taken into account. The control problem therefore cannot be solved by authority specification alone; it requires commit architecture capable of recomputing or re-evaluating admissibility at execution time.

8 The Coordination Economy

8.1 Coordination Cost as the Binding Constraint

The term *coordination economy* is used here to describe a class of productive environment in which the technical means of production are not the primary constraint on value creation, and the central bottleneck lies instead in preserving coherent and admissible action across distributed agents. This environment is characteristic of knowledge-intensive industries, platform ecosystems, regulated organizations, and multi-agent AI systems, where many actors can generate analysis, recommendations, and even execution-ready actions at low marginal cost, but where the difficulty of maintaining aligned direction and valid commitment grows with system scale, interdependence, and speed (Malone & Crowston, 1994; Parker et al., 2016; Puranam et al., 2012).

In this sense, coordination cost should not be understood narrowly as the cost of communication or synchronization. In high-complexity environments, the deeper problem is that action must satisfy two distinct structural conditions at once. First, intent must survive translation into workflow. Strategic and managerial decisions must be decomposed, routed, sequenced, escalated, and reconciled without silently altering what the organization is actually trying to do. Second, a specific act must remain admissible at the moment of execution: the authority, context, and constraint set under which the action becomes irreversible must still render that action valid under current conditions.

The central bottleneck in a coordination economy is therefore not merely moving information or obtaining approvals. It is preserving intent through orchestration while still resolving admissibility at commitment. Where this capability is weak, organizations can generate abundant decisions, analyses, and workflow activity while still failing to realize valid coordinated action.

8.2 From Standing Authority to Orchestrated Action

The earlier sections model legitimacy as a structural component of autonomy: a relatively durable property of recognized standing within an organizational decision system. That concept remains necessary, but it is not sufficient. Possessing standing authority does not ensure that the authority will be translated into action faithfully. Between recognized decision rights and irreversible execution lies an intermediate layer: workflow or orchestration architecture.

This layer determines how decisions are operationalized across handoffs, systems, teams, and increasingly AI agents. It determines how a decision is broken into sub-decisions, how tasks are routed, how dependencies are ordered, how exceptions are escalated, and how conflicting constraints are reconciled. In doing so, it becomes the principal site where coherence is either preserved or lost. The present framework therefore proposes that data problems and execution breakdowns are often downstream observability surfaces of orchestration failure rather than the primary origin of drift.

The coordination economy is therefore not only an economy of alignment across distributed actors; it is an economy of preserving intent as decisions are translated into workflows. A system with clear authority but weak orchestration integrity may remain procedurally orderly while advancing a distorted version of the original objective. Conversely, a system with strong orchestration but weak standing legitimacy may route work efficiently while lacking recognized mandate or acceptance. Durable coordinated performance requires both.

8.3 Why Flow, Resistance, and Heat Matter at Scale

The flow–resistance–heat framework developed in Sections 5 and 6 remains central to this analysis. It models the organizational cost of moving decisions and actions through complex structures. In low-complexity, low-throughput environments, the heat generated by suboptimal architecture may remain modest. As throughput, interdependence, and coupling increase, however, the architecture's resistance profile becomes increasingly consequential. Under the model's assumptions, systems with higher resistance generate more dissipation, greater backlog, and more frequent escalation for the same nominal level of action flow. The result is that architectural advantages compound with scale.

In coordination-economy environments, however, throughput alone is not the relevant measure of success. A system may exhibit high flow in the sense of rapid decision preparation, routing, and authorization, while still failing to preserve the original intent through the orchestration layer. In such cases, rework, escalation, conflict, and fatigue may arise not only from high volume under resistance, but from repeated attempts to force throughput through workflows that already encode a drifted objective. Heat therefore captures not only overload but also dissipation generated by compensating for distorted coordination logic.

For this reason, the present framework distinguishes analytically between *coordination flow* and *commitment*. Coordination flow refers to the movement of intent, information, authority claims, and preparatory decisions through the organizational network. Commitment refers to the moment at which action becomes materially or institutionally irreversible—for example, deployment of code, release of funds, grant of access, issuance of an instruction, or execution of an AI-originated action in a live environment.

This distinction matters because the structural conditions required for high flow are not identical to those required for valid commitment. A system may be optimized for throughput while still committing inadmissible actions if the final execution point does not resolve whether the action is still valid under current conditions. In such cases, earlier governance, documentation, and workflow steps function only as preparation. They may shape the action, but they do not control it in the strong sense unless admissibility is resolved where the system commits.

The coordination economy therefore operates under a dual structural burden. Organizations must reduce resistance sufficiently to sustain coherent flow through the orchestration layer, but they must also ensure that irreversible action is conditioned on execution-time validity. The first burden is captured by the resistance and heat relations introduced earlier. The second burden introduces an additional architectural requirement: the system must not merely transport authority claims; it must determine whether action is still admissible when execution occurs.

8.4 Preparation, Orchestration, Commitment, and Control

A central implication of the present framework is that preparation should not be conflated with control, and that orchestration should not be mistaken for a purely technical implementation layer. Organizations invest heavily in preparatory structures: governance committees, policy documents, approval chains, workflow steps, model documentation, and audit trails. These may shape the generation and routing of action significantly. But unless they preserve intent through orchestration and bind the point at which irreversible action occurs, they do not control execution in the strongest sense.

This distinction matters because many organizational pathologies arise precisely when preparation is mistaken for control. A system may appear well-governed because decisions are documented, authority domains are assigned, workflows are formalized, and approvals are captured, while still allowing either of two structural failures. First, the workflow may have already encoded a distorted version of the original decision. Second, an inadmissible act may still commit because no mechanism resolves current validity at execution time. In such cases, the organization has designed the preparation path and perhaps even the process map, but not the full control path.

The coordination economy therefore depends on three related but non-identical architectures. The first is the *decision architecture* that assigns standing authority. The second is the *orchestration architecture* that preserves coherence, reduces resistance, and translates decisions into coordinated action paths. The third is the *commit architecture* that determines whether the act remains admissible under current conditions at the moment the system commits. Durable advantage requires all three.

8.5 Two Distinct Failure Modes: Orchestration Failure and Commit Failure

This refinement clarifies that large-scale coordination failure can arise through two analytically distinct mechanisms, which may occur independently or jointly.

First, systems may fail through *orchestration failure*. In this case, the original strategic or operational intent is progressively reinterpreted as it moves through decomposition logic, routing rules, handoffs, sequencing structures, escalation paths, vendor interfaces, software systems, and automation pipelines. Each local translation may appear reasonable, and each step may remain procedurally valid, yet the action ultimately committed is no longer aligned with the intent the organization sought to realize. The failure here is not necessarily stale authority but structural drift: the system preserves permissions, process conformance, or local optimization while losing global directional integrity. The first practical instability often appears in the orchestration layer, while execution and data reveal the consequences only later.

Second, systems may fail through *commit failure*. In this case, the intended action may remain coherent and the authority under which it was prepared may initially have been valid, but by the moment of execution the conditions that justified the action no longer hold. The context has changed, the authorization has expired, the risk state has shifted, or a higher-order constraint now blocks execution. The failure here is temporal rather than translational: the problem is not that the action was distorted in transit, but that it was not revalidated where it became irreversible.

These failure modes may occur independently or jointly. A system may preserve orchestration integrity yet execute stale authority. A system may preserve admissibility at execution time yet coordinate around a distorted objective. Or it may do both: drift structurally while also failing to validate commitment. The coordination economy is therefore not reducible to either throughput optimization or compliance checking alone. Its central problem is the joint preservation of coherent intent through orchestration and admissible action at commitment under distributed conditions.

A system may therefore be structurally coherent yet inadmissible at commitment, or admissible at commitment yet globally incoherent with respect to the organization's broader intent.

8.6 Why AI Increases the Importance of Orchestration and Commit Architecture

These issues become particularly acute in AI-augmented organizations. Artificial intelligence can increase the effective rate at which options are generated, evaluated, and advanced toward execution. As argued in Section 9, this can increase potential flow substantially. But AI also compresses the interval between preparation and commitment, scales the number of actions under consideration, and increases the number of translation points through which organizational intent must pass before action occurs.

As a result, AI systems intensify both major coordination burdens. They may increase orchestration risk by introducing additional translation layers between organizational intent and operational action, especially when objective functions, proxy metrics, workflow engines, or learned policies substitute for direct human judgment. They also increase admissibility risk by enabling action at a pace that exceeds the organization's capacity to determine whether authority and context remain valid at the point of execution. Under these conditions, a governance arrangement that remains external to the orchestration and execution layers may shape behavior but cannot reliably control it.

The model therefore predicts that in AI-native coordination economies, competitive advantage will not accrue simply to organizations that generate more decisions, automate more tasks, or distribute more autonomy. It will accrue to organizations that build architectures capable of doing three things simultaneously:

1. preserving the coherence of intent through workflow and orchestration,
2. minimizing structural resistance so that flow remains stable rather than heat-generating, and
3. resolving admissibility at the point of irreversible execution.

8.7 Implications for Organizational Advantage at Scale

Under these conditions, the central unit of competition shifts. It is no longer sufficient to compare organizations by production efficiency, information access, or local decision quality alone. The relevant question becomes whether the organization can preserve valid coordinated action under conditions of high throughput, distributed interdependence, and changing context.

The model therefore predicts that organizations with lower structural resistance, stronger orchestration integrity, and stronger execution-time validity mechanisms will realize compounding advantages at scale. Specifically, it predicts that:

- organizations that expand autonomy without reducing resistance will generate heat rather than proportional value;
- organizations that formalize decision rights without preserving intent through orchestration will exhibit orderly but drifted execution;
- organizations that improve throughput without embedding execution-time admissibility checks will increase the risk of invalid commitments;
- organizations that maintain local validity while allowing intent to fragment through workflow layers will exhibit globally misaligned but procedurally correct execution;
- organizations that jointly optimize decision architecture, orchestration architecture, resistance reduction, and commit architecture will be able to sustain higher effective throughput before encountering instability thresholds.

The coordination economy, as defined here, therefore refers not merely to an environment in which coordination costs are high. It refers to an environment in which value creation increasingly depends on an organization's ability to preserve coherent intent through orchestration while also resolving admissibility at the point of irreversible execution. In such environments, resistance management, orchestration integrity, network conservation, and commit-time admissibility become coequal determinants of organizational viability.

9 Limitations of the Model

The following limitations are inherent to the framework as currently formulated and should be addressed in any research program designed to evaluate its claims.

1. **Unvalidated measurement:** The model's constructs—Legitimacy, Capability, Insignificance, Risk, and Resistance—are defined conceptually but do not yet have validated psychometric or organizational measurement instruments. The algebraic precision of the expressions is not currently matched by corresponding precision in operationalization. Construct validation studies are a necessary precondition for empirical testing of any of the predictions in Section 12.

2. **Assumed functional forms:** The specific functional forms—multiplicative for Autonomy, linear subtraction for Legitimacy and Capability, ratio for Flow, and quadratic for Heat—are modeling choices made for analytical tractability and conceptual coherence. Empirical relationships among the underlying variables may be nonlinear, discontinuous, threshold-dependent, or interaction-dependent in ways that these forms do not capture. The model makes no claim that these are the uniquely correct functional forms.

3. **Static structure:** The framework describes system state at a point in time. Organizational systems are dynamic: resistance changes as architecture adapts, legitimacy changes as relationships develop, and capability changes through learning and attrition. The model does not currently specify differential equations or feedback dynamics governing these changes.

4. **Interpersonal and cultural factors:** The framework models organizations as flow networks and does not represent interpersonal relationships, organizational culture, or historical path dependence— all of which are empirically significant (Schein, 1985). The model's resistance variable is intended to absorb some cultural effects, but this substitution is not theoretically grounded.

5. **Kirchhoff analogy limitations:** Kirchhoff's laws are formally valid for linear, time-invariant networks. Organizational decision networks are nonlinear, time-varying, and subject to exogenous shocks. The conservation constraints introduced in Section 7 should be understood as qualitative structural hypotheses rather than precise quantitative laws.

6. **Endogeneity:** The model specifies a causal structure in which architecture determines resistance, which determines flow and heat. In practice, heat outcomes (conflict, burnout, turnover) alter architectural properties, creating feedback loops not

currently represented in the model. Causal identification in any empirical test will need to address this endogeneity through longitudinal or experimental designs.

7. **Admissibility not yet formalized algebraically:** While the revised framework distinguishes standing legitimacy from execution-time admissibility, it does not yet provide a full algebraic treatment of admissibility comparable to the treatments of Autonomy, Flow, and Heat. The current paper therefore identifies admissibility as a necessary construct and commit architecture as a necessary design layer, but stops short of specifying a complete functional form for how contextual change, temporal decay, and commit-point validation interact.

8. **The "heat" metaphor and reification risk:** The term organizational heat carries strong inferential connotations from physics that may not transfer reliably to organizational contexts. Readers and researchers should treat the term as a label for a proposed latent variable—unproductive dissipation under structural load—rather than as a claim that organizations literally obey thermodynamic laws.

10 Conclusion

This paper has proposed a structural model of organizational dynamics grounded in physical flow analogies and positioned within established theoretical traditions in psychology, systems theory, and organization design. The core framework formalizes Autonomy as the product of Legitimacy and Capability, Action Flow as the ratio of Autonomy to Complexity Resistance, and Organizational Heat as a quadratic function of Flow and Resistance. These are modeling hypotheses—deliberately stated in formal terms to enable testable predictions—not summaries of existing findings.

The framework is motivated by, but distinct from, its theoretical antecedents. Self-determination theory, cognitive load theory, cybernetics, normal accident theory, and coordination economics each supply evidence for one or more of the model's conceptual components; none of them directly implies the specific algebraic forms proposed here.

The model's primary theoretical contributions are fourfold. First, it proposes a unified formal language for phenomena—learned helplessness, burnout, coordination failure, escalation overload—that are typically treated in separate literatures. Second, it applies Kirchhoff-like conservation constraints to decision networks, generating structural predictions about where coordination failures will occur and why they are stable. Third, it distinguishes standing legitimacy from execution-time admissibility, thereby separating the problem of coordinated preparation from the problem of valid commitment. Fourth, it applies the flow–resistance–heat framework to AI-augmented organizations, generating predictions about the conditions under which AI-driven capability expansion produces organizational instability or invalid commitment rather than productivity.

The testable predictions developed in Section 12 provide a concrete research agenda. Testing them would require both construct validation—developing reliable measures for the model's latent variables—and structural equation modeling, longitudinal designs, or experimental methods capable of separating coherence failure from admissibility failure and estimating the functional forms proposed. Until such tests are conducted, the framework should be understood as a structured theoretical proposal whose value lies in the precision and falsifiability of its claims, not in their prior empirical confirmation.

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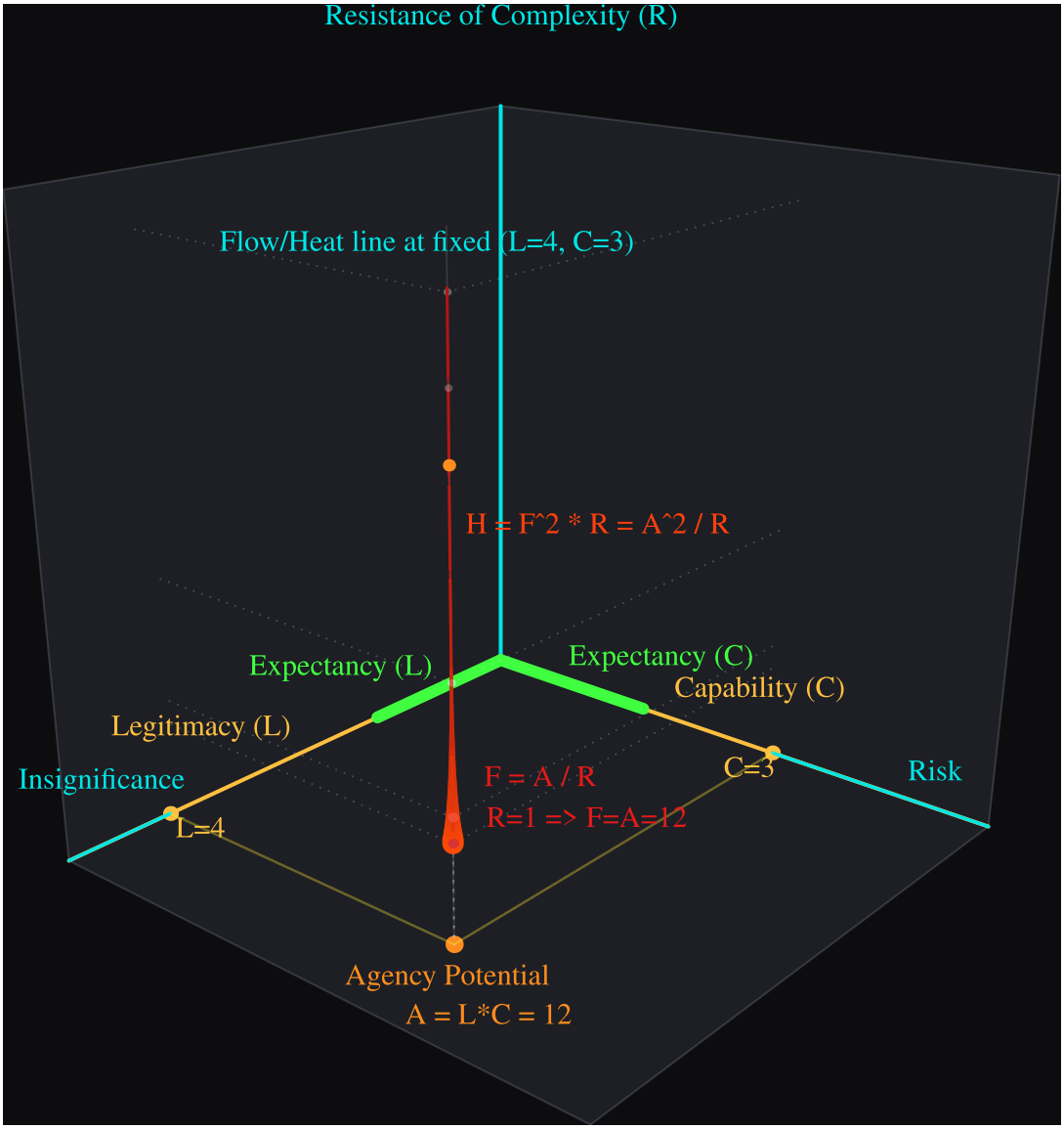

Figure 1: Integrated diagram of the Autonomy–Flow–Heat framework. Arrows denote the causal dependencies proposed by the model. The structure mirrors an electrical circuit with a driving potential and resistive dissipation, but the mapping is analogical; the diagram is not intended to imply that organizational quantities are measurable in physical units.